Introduction to Robotics

Lecture 3: Orientation Representations and Exponential Coordinates

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Template made by Panagiotis Papagiannopoulos

Logistics

Laboratory of Automation & Robotics

- **Lectures**:
  - 12 Lectures
  - 1 Recitation Lecture
  - Lab
- **Examination**:
  - 4 Homeworks **(30% of total grade)**
  - Oral Exam **(50% of total grade)**
  - Lab **(20% of total grade)**
- **Office Hours**:
  - **Tue-Thu** *(10:00-12:00)*
  - 24/7 by email (`costashatz@upatras.gr`, subject: *ECE_IR_AM*)
- **Material and Announcements**

What is this course about?

All the basics for delving into the field of Robotics:

- Rigid Body Motions in 2D and 3D
- Homogeneous Transformations and Exponential Coordinates
- Forward Kinematics
- Inverse Kinematics
- Jacobians and Singularities
- Velocity Control
- Trajectory Generation
- Dynamics of a Single Rigid Body and Open-Chains
- Force/Torque Control

Lectures/Lab Format

- **3-hour lectures:**
  - Theory
  - Hints for personal reading!
  - Live code examples!
  - Ask questions please!
- **2-hour labs (in teams):**
  - **20% of total grade**!
  - ROS
  - Arduino
  - Work on real systems!

Skew Symmetric Matrices

The **skew symmetric matrix** representation of a 3-D vector $\boldsymbol{v} = [v_1 \ v_2 \ v_3]^T$ is:

$$\hat{\boldsymbol{v}} = [\boldsymbol{v}] = \boldsymbol{v}_\times = \begin{bmatrix} 0 & -v_3 & v_2 \\ v_3 & 0 & -v_1 \\ -v_2 & v_1 & 0 \end{bmatrix} \in \mathbb{R}^{3\times 3}$$

Many names in the literature:

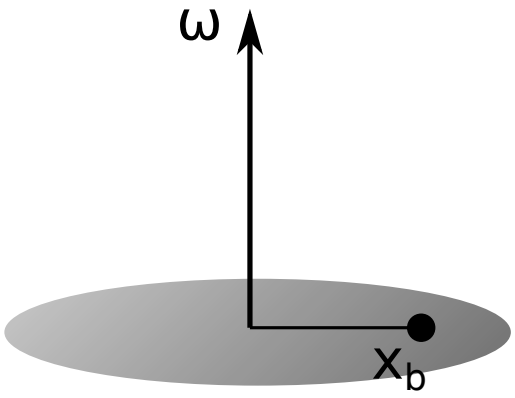
- hat map or matrix, $\hat{\boldsymbol{v}}$
- skew symmetric matrix, $[\boldsymbol{v}]$
- cross product matrix, $\boldsymbol{v}_\times$ or $[\boldsymbol{v}]_\times$

We define $\check{\boldsymbol{v}}_\times = [v_1 \ v_2 \ v_3]^T$ as the inverse operator.

Rotation Kinematics (aka velocities)

$$\boldsymbol{x}_w = \boldsymbol{R}_{wb}\boldsymbol{x}_b$$

$\boldsymbol{x}_b$ is static. **How can we compute** $\dot{\boldsymbol{x}}_w$**?**



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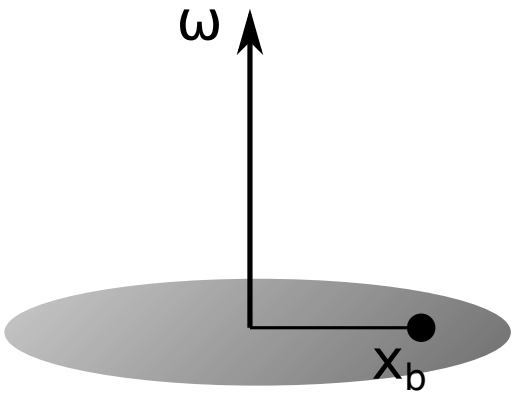
$$\begin{aligned}\dot{\boldsymbol{x}}_w &= \boldsymbol{\omega}_w \times \boldsymbol{x}_w \\ &= \boldsymbol{R}_{wb}(\boldsymbol{\omega}_b \times \boldsymbol{x}_b)\end{aligned}$$



How else?

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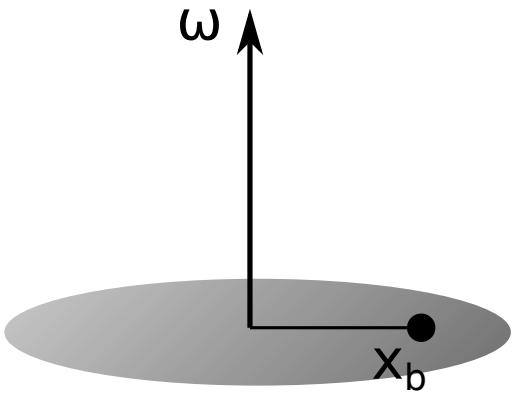

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How else?

$$\begin{aligned}\boldsymbol{x}_w &= \boldsymbol{R}_{wb}\boldsymbol{x}_b \Rightarrow \\ \dot{\boldsymbol{x}}_w &= \dot{\boldsymbol{R}}_{wb}\boldsymbol{x}_b + \boldsymbol{R}_{wb}\dot{\boldsymbol{x}}_b \\ \dot{\boldsymbol{x}}_w &= \dot{\boldsymbol{R}}_{wb}\boldsymbol{x}_b, \text{ since } \dot{\boldsymbol{x}}_b = 0\end{aligned}$$

So, what we do have?

Rotation Kinematics (aka velocities)

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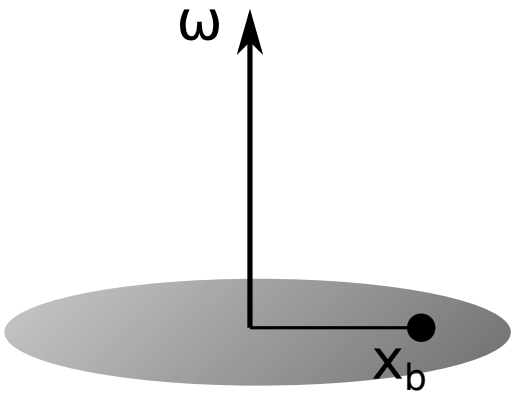

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So, what we do have?

$$\begin{aligned}\dot{\boldsymbol{R}}_{wb}\boldsymbol{x}_b = \boldsymbol{R}_{wb}(\boldsymbol{\omega}_b \times \boldsymbol{x}_b) = \boldsymbol{R}_{wb}\hat{\boldsymbol{\omega}}_b\boldsymbol{x}_b \Rightarrow \\ \dot{\boldsymbol{R}}_{wb} = \boldsymbol{R}_{wb}\hat{\boldsymbol{\omega}}_b\end{aligned}$$

Rotation Kinematics (2)

Let's appreciate the previous result:

$$\dot{\boldsymbol{R}}_{wb} = \boldsymbol{R}_{wb}\hat{\boldsymbol{\omega}}_b$$

What does this mean? What can I do with it?

Rotation Kinematics (2)

Let's appreciate the previous result:

$$\dot{\boldsymbol{R}}_{wb} = \boldsymbol{R}_{wb}\hat{\boldsymbol{\omega}}_b$$

What does this mean? What can I do with it?

It means that if I have a measurement of the angular velocity ω_b, I can use it to integrate the full rotation matrix!

In other words:

- I can do forward simulation
- I can use it in combination with gyros/IMUs for state estimation/odometry

With some math manipulation, we can also have ("world space angular velocity"):

$$\dot{\boldsymbol{R}}_{wb} = \hat{\boldsymbol{\omega}}_w\boldsymbol{R}_{wb}$$

Angular Velocities

Let's summarize:

$$\hat{\omega}_w = \dot{\boldsymbol{R}}_{wb}\boldsymbol{R}_{wb}^T$$
$$\hat{\omega}_b = \boldsymbol{R}_{wb}^T\dot{\boldsymbol{R}}_{wb}$$

- $\boldsymbol{\omega}_w \in \mathbb{R}^3$ is the angular velocity of the body with respect to (wrt) the world frame
- $\boldsymbol{\omega}_b \in \mathbb{R}^3$ is the angular velocity of the body wrt the body frame
- $\boldsymbol{\omega}_b$ **IS NOT** the angular velocity relative to the moving body frame, but relative to the fixed world frame
- $\boldsymbol{\omega}_w$ **DOES NOT** depend on the choice of the body frame
- $\boldsymbol{\omega}_b$ **DOES NOT** depend on the choice of the world frame
- $\dot{\boldsymbol{R}}_{wb}\boldsymbol{R}_{wb}^T$ is independent from the body frame
- $\boldsymbol{R}_{wb}^T\dot{\boldsymbol{R}}_{wb}$ is independent from the world frame
- Let's assume an angular velocity expressed in frame $\{d\}$, $\boldsymbol{\omega}_d$, we can express it in the frame $\{c\}$: $\boldsymbol{\omega}_c = \boldsymbol{R}_{cd}\boldsymbol{\omega}_d$

How can we represent orientations?

How can we represent orientations?

- Rotation Matrices

How can we represent orientations?

- Rotation Matrices
- Euler Angles
- Axis-Angle
- Quaternions

Euler Angles

Euler Angles:

$$\begin{aligned} \boldsymbol{R} &= \boldsymbol{R}_1 \boldsymbol{R}_2 \boldsymbol{R}_3 \\ &= \underbrace{\boldsymbol{R}_{\alpha_1}(\theta_1)}_{\boldsymbol{R}_1} \underbrace{\boldsymbol{R}_{\alpha_2}(\theta_2)}_{\boldsymbol{R}_2} \underbrace{\boldsymbol{R}_{\alpha_3}(\theta_3)}_{\boldsymbol{R}_3} \end{aligned}$$

where α_i are the axes of rotation and θ_i the corresponding angles of rotation.

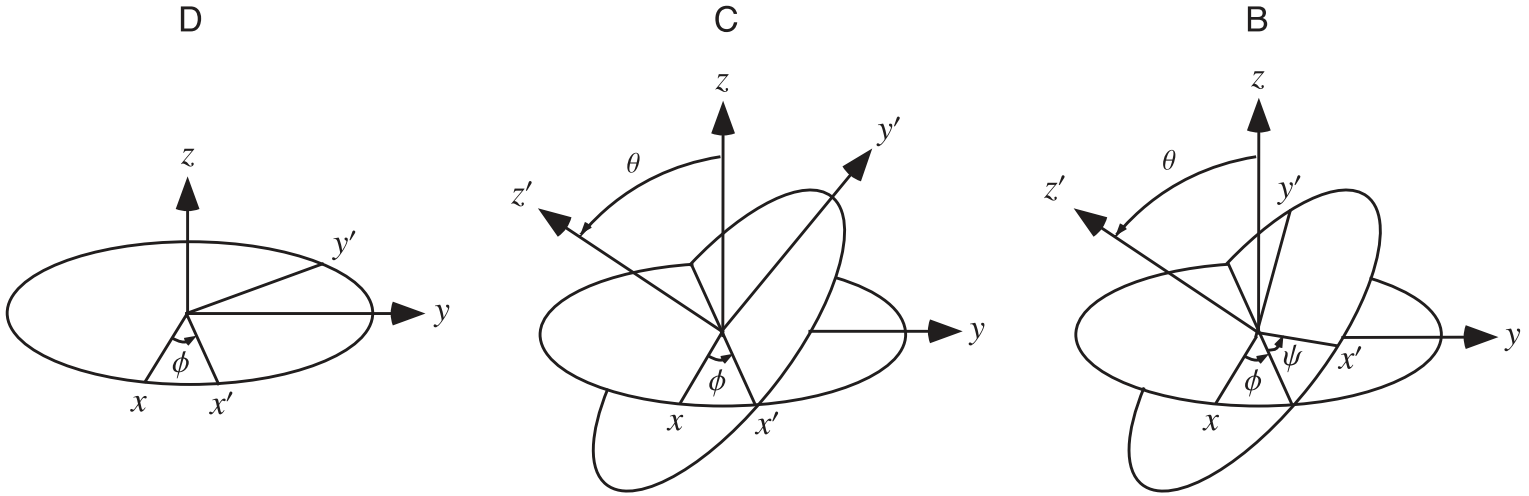

Figure: Taken from WolframAlpha

Euler Angles (2)

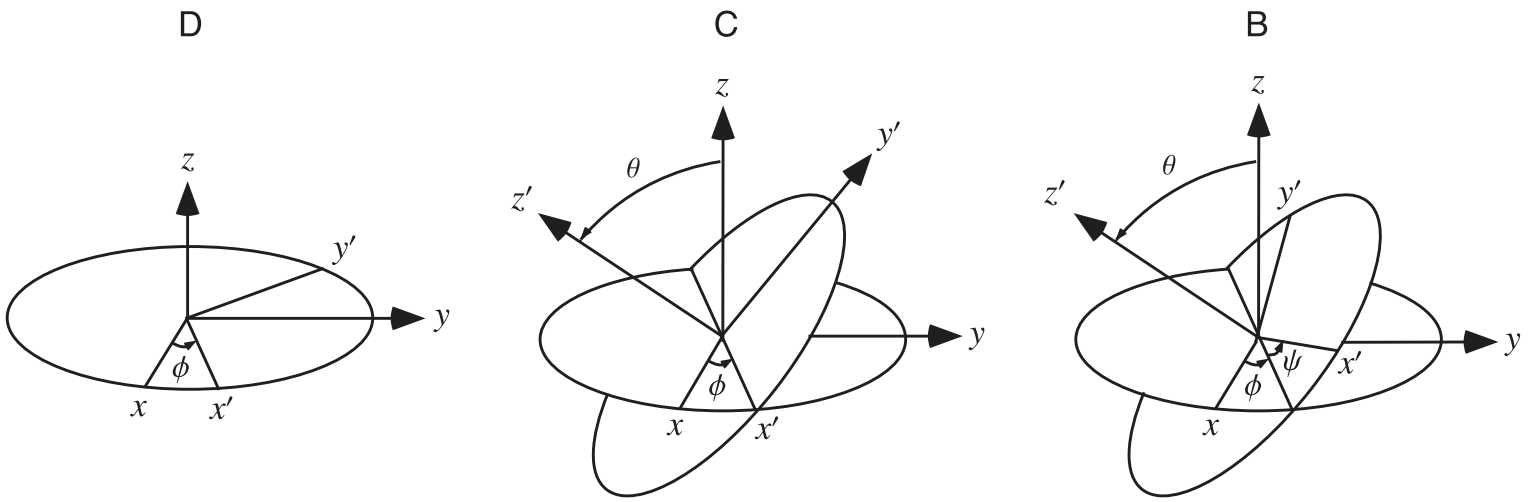

Figure: Taken from WolframAlpha

- There are many conventions: *ZYX* (yaw-pitch-roll) and *XYZ* (roll-pitch-yaw) the most popular ones
  - $\theta_1 \in [-\pi, \pi], \theta_2 \in [-\frac{\pi}{2}, \frac{\pi}{2}], \theta_3 \in [-\pi, \pi]$
  - **Singularity** at $\theta = \pm\frac{\pi}{2}$
- We can convert from/to rotation matrices

Euler Angles and Angular Velocity

ZYX Euler Angles:

$$\boldsymbol{\omega}_b = \begin{bmatrix} -\sin\theta_2 & 0 & 1 \\ \cos\theta_2 \sin\theta_3 & \cos\theta_3 & 0 \\ \cos\theta_3 \cos\theta_2 & -sin\theta_3 & 0 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

$$\boldsymbol{\omega}_w = \begin{bmatrix} 0 & -\sin\theta_1 & \cos\theta_2 \cos\theta_1 \\ 0 & \cos\theta_1 & \cos\theta_2 \sin\theta_1 \\ 1 & 0 & -\sin\theta_2 \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix}$$

XYZ Euler Angles:

$$\vdots$$

$$\vdots$$

Axis-Angle or Exponential Coordinates

Axis-Angle:

$$\boldsymbol{\theta} = \boldsymbol{r}\theta$$

where $\boldsymbol{r} \in \mathbb{R}^3$ is the "axis of rotation" unit vector, and $\theta \in \mathbb{R}$ is the rotation angle.

- **Singularity** at $\theta = \pm\pi$
- $\boldsymbol{\theta}_{wb} = \widetilde{\log \boldsymbol{R}_{wb}}$
- $\boldsymbol{R}_{wb} = \exp(\widehat{\boldsymbol{\theta}_{wb}})$

Axis-Angle or Exponential Coordinates

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- **Singularity** at $\theta = \pm\pi$
- $\boldsymbol{\theta}_{wb} = \widetilde{\log \boldsymbol{R}_{wb}}$
- $\boldsymbol{R}_{wb} = \exp(\widehat{\boldsymbol{\theta}_{wb}})$
- We can compute the above more efficiently than regular exp/log matrix operations

Axis-Angle or Exponential Coordinates (2)

$$
\begin{aligned}
\exp(\hat{\boldsymbol{\theta}}) = \exp(\theta\hat{\boldsymbol{r}}) &= \sum_{n=0}^{\infty} \frac{1}{n!}(\hat{\boldsymbol{\theta}})^n \\
&= \mathbf{1} + \theta\hat{\boldsymbol{r}} + \frac{1}{2!}\theta^2\hat{\boldsymbol{r}}\hat{\boldsymbol{r}} + \cdots \\
&= \cdots \\
&= \boldsymbol{I}\cos\theta + (1-\cos\theta)\boldsymbol{r}\boldsymbol{r}^T + \sin\theta\hat{\boldsymbol{r}} \\
&= \boldsymbol{I} + \sin\theta\hat{\boldsymbol{r}} + (1-\cos\theta)\hat{\boldsymbol{r}}\hat{\boldsymbol{r}} \text{ (Rodrigues formula)} \\
&= \boldsymbol{R}
\end{aligned}
$$

Similarly:[1]

$$
\widehat{\log(\boldsymbol{R})} = \boldsymbol{\theta} = \theta\boldsymbol{r}
$$

$$
\hat{\boldsymbol{r}} = \frac{1}{2\sin\theta}(\boldsymbol{R} - \boldsymbol{R}^T), \theta = \cos^{-1}\left(\frac{\mathrm{tr}(\boldsymbol{R}) - 1}{2}\right)
$$

[1] We also need to take care of some special cases!

Axis-Angle and Angular Velocity

Axis-Angle:

$$\boldsymbol{\theta} = \boldsymbol{r}\theta$$

$$\boldsymbol{R} = \exp(\widehat{\boldsymbol{\theta}_{wb}})$$

$$= \boldsymbol{I} + \sin\theta\hat{\boldsymbol{r}} + (1-\cos\theta)\hat{\boldsymbol{r}}^2, \ \textit{(Rodrigues formula)} \Rightarrow$$

$$\dot{\boldsymbol{R}} = \dot{\theta}\cos\theta\hat{\boldsymbol{r}} + \sin\theta\dot{\hat{\boldsymbol{r}}} + \dot{\theta}\sin\theta\hat{\boldsymbol{r}}^2 + (1-\cos\theta)(\dot{\hat{\boldsymbol{r}}}\hat{\boldsymbol{r}} + \hat{\boldsymbol{r}}\dot{\hat{\boldsymbol{r}}}) \Rightarrow$$

$$\dots$$

$$\Rightarrow \boldsymbol{\omega}_w = \dot{\theta}\boldsymbol{r} + \sin\theta\dot{\boldsymbol{r}} + \hat{\boldsymbol{r}}\dot{\boldsymbol{r}}(1-\cos\theta)$$

$$\Rightarrow \boldsymbol{\omega}_b = \dot{\theta}\boldsymbol{r} + \sin\theta\dot{\boldsymbol{r}} - \hat{\boldsymbol{r}}\dot{\boldsymbol{r}}(1-\cos\theta)$$

Quaternions

Let's start with the **Axis-Angle:**

$$\boldsymbol{\theta} = \boldsymbol{r}\theta$$

A **unit quaternion** describing the same orientation/rotation is given by:

$$\boldsymbol{q} = \begin{bmatrix} \cos\frac{\theta}{2} \\ \boldsymbol{r}\sin\frac{\theta}{2} \end{bmatrix} \in \mathbb{R}^4$$

- $\cos\frac{\theta}{2}$ is the **scalar** part, and $\boldsymbol{r}\sin\frac{\theta}{2}$ is the **vector** part
- For $\boldsymbol{q}$ to be a valid orientation/rotation, we also need $\|\boldsymbol{q}\| = 1$
- $\boldsymbol{q}$ and $-\boldsymbol{q}$ describe the **same orientation**, but *different rotations!* (double cover)

Quaternion Math

- $\boldsymbol{q}_{wb}$ is a rotation from "body frame" to "world frame"
- $\boldsymbol{q}_{wa} = \boldsymbol{q}_{wb} \odot \boldsymbol{q}_{ba}$

$$\boldsymbol{q}_{wa} = \begin{bmatrix} s_{wb} \\ \boldsymbol{v}_{wb} \end{bmatrix} \odot \begin{bmatrix} s_{ba} \\ \boldsymbol{v}_{ba} \end{bmatrix} = \begin{bmatrix} s_{wb} s_{ba} - \boldsymbol{v}_{wb}^T \boldsymbol{v}_{ba} \\ s_{wb} \boldsymbol{v}_{ba} + s_{ba} \boldsymbol{v}_{wb} + \boldsymbol{v}_{wb} \times \boldsymbol{v}_{ba} \end{bmatrix}$$

- **Quaternion Conjugate** (Inverse):

$$\boldsymbol{q}_{wb}^{\dagger} = \begin{bmatrix} s_{wb} \\ -\boldsymbol{v}_{wb} \end{bmatrix}$$

- $\boldsymbol{q}_{wb}^{\dagger} \odot \boldsymbol{q}_{wb} = \boldsymbol{q}_0$, $\boldsymbol{q}_0 = \begin{bmatrix} 1 \\ \boldsymbol{0} \end{bmatrix}$
- $\begin{bmatrix} 0 \\ \boldsymbol{x}_w \end{bmatrix} = \boldsymbol{q}_{wb} \odot \begin{bmatrix} 0 \\ \boldsymbol{x}_b \end{bmatrix} \odot \boldsymbol{q}_{wb}^{\dagger}$, *(rotating a vector/point)*

Quaternions and Angular Velocity

Time derivative of Quaternion:

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{q} \odot \begin{bmatrix} 0 \\ \boldsymbol{\omega}_b \end{bmatrix}$$

- We can now integrate quaternions

Quaternions and Angular Velocity

Time derivative of Quaternion:

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{q} \odot \begin{bmatrix} 0 \\ \boldsymbol{\omega}_b \end{bmatrix}$$

- We can now integrate quaternions
- **Beware!** Integrating naively with $\dot{\boldsymbol{q}}$ (e.g. with Euler) results in a non unit-norm quaternion!

Quaternions and Angular Velocity

Time derivative of Quaternion:

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{q} \odot \begin{bmatrix} 0 \\ \boldsymbol{\omega}_b \end{bmatrix}$$

- We can now integrate quaternions
- **Beware!** Integrating naively with $\dot{\boldsymbol{q}}$ (e.g. with Euler) results in a non unit-norm quaternion!
- **Lots of "custom" functions to use the quaternions!**

Quaternion Magic

$$\boldsymbol{q}_{wa} = \begin{bmatrix} s_{wb} \\ \boldsymbol{v}_{wb} \end{bmatrix} \odot \begin{bmatrix} s_{ba} \\ \boldsymbol{v}_{ba} \end{bmatrix} = \begin{bmatrix} s_{wb}s_{ba} - \boldsymbol{v}_{wb}^T \boldsymbol{v}_{ba} \\ s_{wb}\boldsymbol{v}_{ba} + s_{ba}\boldsymbol{v}_{wb} + \boldsymbol{v}_{wb} \times \boldsymbol{v}_{ba} \end{bmatrix}$$

How can we write the above such that we do it via matrix multiplication?

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$$\boldsymbol{q}_{wa} = \begin{bmatrix} s_{wb} \\ \boldsymbol{v}_{wb} \end{bmatrix} \odot \begin{bmatrix} s_{ba} \\ \boldsymbol{v}_{ba} \end{bmatrix} = \underbrace{\begin{bmatrix} s_{wb} & -\boldsymbol{v}_{wb}^T \\ \boldsymbol{v}_{wb} & s_{wb}\boldsymbol{I} + \hat{\boldsymbol{v}}_{wb} \end{bmatrix}}_{L(\boldsymbol{q}_{wb})} \begin{bmatrix} s_{ba} \\ \boldsymbol{v}_{ba} \end{bmatrix}$$

$$= \underbrace{\begin{bmatrix} s_{ba} & -\boldsymbol{v}_{ba}^T \\ \boldsymbol{v}_{ba} & s_{ba}\boldsymbol{I} - \hat{\boldsymbol{v}}_{ba} \end{bmatrix}}_{R(\boldsymbol{q}_{ba})} \begin{bmatrix} s_{wb} \\ \boldsymbol{v}_{wb} \end{bmatrix}$$

Quaternion Magic (2)

A few interesting relations:

$$\begin{aligned} \boldsymbol{q}_{wb} \odot \boldsymbol{q}_{ba} &= L(\boldsymbol{q}_{wb})\boldsymbol{q}_{ba} \\ &= R(\boldsymbol{q}_{ba})\boldsymbol{q}_{wb} \end{aligned}$$

$$\begin{aligned} L(\boldsymbol{q}^{\dagger}) &= L(\boldsymbol{q})^{T} = L(\boldsymbol{q})^{-1} \\ R(\boldsymbol{q}^{\dagger}) &= R(\boldsymbol{q})^{T} = R(\boldsymbol{q})^{-1} \end{aligned}$$

Quaternion Magic (3)

We can use the same logic in all quaternion operations:

- $\boldsymbol{q}^\dagger = \begin{bmatrix} s \\ -\boldsymbol{v} \end{bmatrix} = \boldsymbol{T}\boldsymbol{q} = \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & -\boldsymbol{I} \end{bmatrix}}_{\boldsymbol{T}} \begin{bmatrix} s \\ \boldsymbol{v} \end{bmatrix}$
- Rotate a vector/point:

$$\begin{bmatrix} 0 \\ \boldsymbol{x}_w \end{bmatrix} = \boldsymbol{q}_{wb} \odot \begin{bmatrix} 0 \\ \boldsymbol{x}_b \end{bmatrix} \odot \boldsymbol{q}_{wb}^\dagger$$

$$\underbrace{\begin{bmatrix} \boldsymbol{0} \\ \boldsymbol{I} \end{bmatrix}}_{\boldsymbol{H} \in \mathbb{R}^{4\times 3}} \boldsymbol{x}_w = \boldsymbol{q}_{wb} \odot \boldsymbol{H}\boldsymbol{x}_b \odot \boldsymbol{q}_{wb}^\dagger$$

$$\boldsymbol{H}\boldsymbol{x}_w = L(\boldsymbol{q}_{wb})(R(\boldsymbol{q}_{wb})^T \boldsymbol{H}\boldsymbol{x}_b)$$

$$\boldsymbol{x}_w = \underbrace{\boldsymbol{H}^T L(\boldsymbol{q}_{wb}) R(\boldsymbol{q}_{wb})^T \boldsymbol{H}}_{\boldsymbol{R}_{wb}} \boldsymbol{x}_b$$

Quaternion Magic (4)

We can use the same logic in all quaternion operations:

$$\vdots$$

$$\dot{\boldsymbol{q}} = \frac{1}{2}\boldsymbol{q} \odot \begin{bmatrix} 0 \\ \boldsymbol{\omega}_b \end{bmatrix}$$

$$= \frac{1}{2} L(\boldsymbol{q}) \boldsymbol{H} \boldsymbol{\omega}_b$$

Rodrigues Parameters and Quaternions

Rodrigues Parameters:[2]

$$\boldsymbol{\theta} = \boldsymbol{r} \tan \frac{\theta}{2} \in \mathbb{R}^3$$

where $\boldsymbol{r} \in \mathbb{R}^3$ is the "axis of rotation" unit vector, and $\theta \in \mathbb{R}$ is the rotation angle. $\boldsymbol{\theta}$ is referred to as "Rodrigues" or "Gibbs" vector.

Quaternions and Rodrigues Parameters:

$$\boldsymbol{q} = \varphi(\boldsymbol{\theta}) = \frac{1}{\sqrt{1 + \boldsymbol{\theta}^T \boldsymbol{\theta}}} \left(\begin{bmatrix} 1 \\ \boldsymbol{\theta} \end{bmatrix} \right)$$

$$\boldsymbol{\theta} = \varphi^{-1}(\boldsymbol{q}) = \frac{\boldsymbol{v}}{s}$$

[2] https://en.wikipedia.org/wiki/Rotation_formalisms_in_three_dimensions

Orientation Representations - Code Example

jupyter 2-orientations Last Checkpoint: 4 minutes ago

File Edit View Run Kernel Settings Help

Code

```
theta = 1.
axis = np.array([[0., 0., 1.]]).T

# quaternion
q = np.block([[np.cos(theta/2.)], [axis * np.sin(theta/2)]])

# rotation matrix
R = quat_to_rotation_matrix(q)

omega_b = np.array([[1., 0., 0.]]).T
```

Bibliography

Chapter 3 from **Modern Robotics: Mechanics, Planning, and Control**, *Kevin M. Lynch and Frank C. Park*, 2017, Cambridge University Press. ebook

Thank you

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